Mohamed Ali Saber

Mechatronics Engineer — Industrial Automation & Control

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LinkedIn — Portfolio — Cairo, Egypt

Professional Summary

Mechatronics Engineer focused on **Industrial Automation and Control Systems**, with hands-on experience using **Siemens S7-1200 PLC and TIA Portal**. Experienced in **Ladder Logic, SCL, HMI, SCADA, WinCC, PROFINET, Modbus, VFDs, Soft Starters, motor control, sensors, and PID control**. Strong foundation in sequential control, electrical control circuits, and industrial troubleshooting.

Education

Bachelor of Mechatronics Engineering — Mansoura University **GPA: 3.66 / 4.00**

Industrial Automation Skills

- **PLC:** Siemens S7-1200, TIA Portal, Ladder Logic, SCL, PLC Diagnostics
- **HMI / SCADA:** WinCC, HMI Design, Tags, Alarms, Process Monitoring
- **Industrial Communication:** PROFINET, Modbus RTU/TCP
- **Motor Control:** VFD, Soft Starter, Star-Delta, Forward/Reverse
- **Control Systems:** PID, ON/OFF Control, Feedback Systems
- **Electrical Control:** Relays, Contactors, Timers, Interlocking
- **Industrial Sensors:** Proximity, Photoelectric, Limit Switches, Encoders, Temperature, Pressure
- **Simulation:** Factory I/O

Projects

Automated Color Mixing System *Siemens S7-1200 • TIA Portal • SCL • HMI/SCADA*

Developed an automated color mixing system where the operator selects the desired color through the HMI/SCADA interface. Empty containers are transported by a conveyor and filled with primary colors according to predefined ratios. Implemented the complete control sequence, including conveyor operation, color dosing, and mixing, using SCL on a Siemens S7-1200 PLC.

Micromouse Autonomous Robot *PID Control • Encoders • IMU • Flood-Fill Navigation*

Developed an autonomous maze-solving robot using PID control and encoder feedback for accurate motion control. Integrated IMU sensors for orientation and implemented the Flood-Fill algorithm for autonomous maze navigation.

CrashXpert — IoT Accident Detection System *ESP32 • Accelerometer • GPS • MQTT*

Developed an IoT-based vehicle accident detection system using an ESP32, accelerometer, and GPS sensors. The system detects potential accidents, determines the vehicle's location, and transmits accident data through MQTT for real-time monitoring and emergency response.

Experience

Head of Software Circle — Momentum MU **Sept 2023 – Sept 2024**

Led embedded and software development for robotics projects and coordinated integration between control, sensors, and mechanical systems.

Software Instructor — Luminous MU

Mentored students in embedded systems, robotics programming, and control fundamentals.

Technical Skills

- **Programming:** C, C#, Python
- **Embedded:** STM32, ESP32, Arduino
- **Software:** TIA Portal, WinCC, Factory I/O, Git, GitHub
- **Communication:** PROFINET, Modbus RTU/TCP, UART, SPI, I2C, MQTT

Achievements

- **1st Place** — IEEE IoT Competition
- **1st Place** — IEEE Fire Fighting Robotics Competition
- **2nd Place** — IEEE Micromouse Competition

Training

- Siemens PLC Programming — TIA Portal, LAD & SCL
- Industrial Automation — HMI, SCADA, VFD, Soft Starter & Industrial Communication
- Classic Control
- Embedded Systems

Languages

- Arabic — Native
- English — Intermediate